

INPUT



Text Instruction

Replace “adventure” with “journey”. / Change the emotion to angry. / Convert the voice of the source speech to match the speaker in the reference speech.



Source Speech



Reference Speech for Speaker Editing

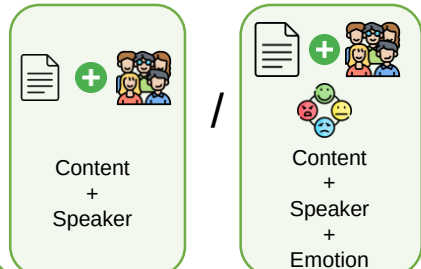
TASK HIERARCHY

Atomic Editing



Compositional Editing

Two-way/Three-way Combinations



ANCHOR

Modification Anchors



Text span target (e.g., “journey”)

Speaker target

Emotion label (e.g., angry)

Style label (e.g., public broadcast)

Prosody target (e.g., speed)

Paralinguistic event (e.g., sigh)

Acoustic target (e.g., remove noise)

Preservation Anchors



Content (whether the content is preserved in unmodified regions)

Evaluation

Target Success



How well the edit targets are achieved

Preservation Success



How well the unrelated attributes are preserved

Joint Success



Overall success of edits with both aspects